RESEARCH

Open Access

Performance assessment of a puf integrated anaerobic bioreactor for textile effluent decolourization along with lab to pilot scale transition and toxicity assessment

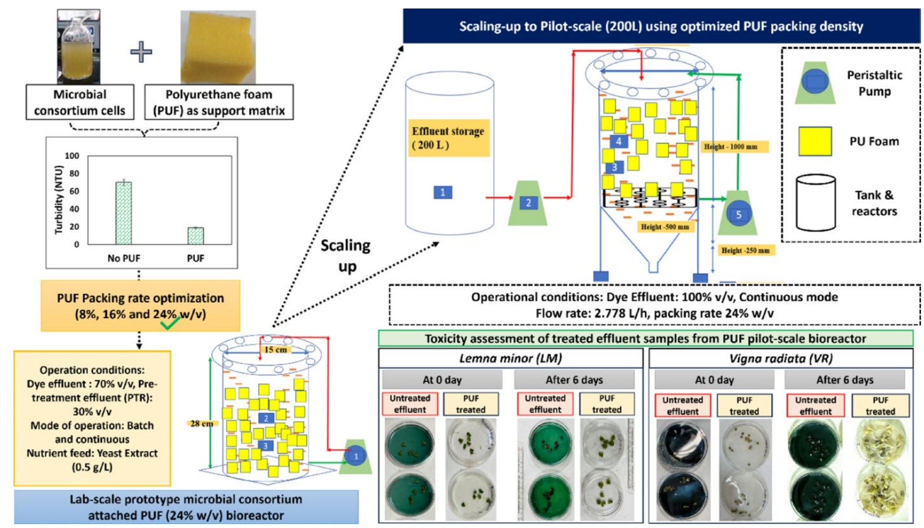
Aman Chaudhary[1,2], Nagesh Vikram Singh[1], Saurabh Samuchiwal[1,4], Vivek Kumar Nair[1,2], Bhupendra Singh Butola[3] and Anushree Malik[1*]

*Correspondence:
Anushree Malik
anushree@rdat.iitd.ac.in
[1]Present address: Applied Microbiology Lab, Centre for Rural Development and Technology, Indian Institute of Technology Delhi, Hauz Khas, New Delhi 110016, India
[2]Present address: School of Interdisciplinary Research, Indian Institute of Technology Delhi, Hauz Khas, New Delhi 110016, India
[3]Department of Textile and Fibre Engineering, Indian Institute of Technology Delhi, Hauz Khas, New Delhi 110016, India
[4]Present address: Department of Biological Sciences and Engineering, Netaji Subhash University of Technology, Dwarka, New Delhi 110078, India

Abstract
Biological remediation of dye-rich textile effluents remains challenging, as these compounds exert toxic, mutagenic, and carcinogenic impacts on humans and aquatic organisms. To address this, a polyurethane foam (PUF) integrated anaerobic bioreactor was designed and subsequently scaled up to facilitate textile effluent decolourization and detoxification. The lab-scale bioreactor optimized with 24% (w/v) PUF packing rate exhibited superior performance over suspended growth bioreactors, achieving 85.2 ± 9.5% color and ~ 73% turbidity removal within 72 h. Furthermore, successful scale-up to a 200 L pilot-scale bioreactor operated continuously for 71 days under real industrial conditions validated the system's robustness, sustaining color and COD removal efficiencies of 76 ± 6.7% and 71.9 ± 4.5%, respectively. Scanning electron microscopy (SEM) confirmed dense microbial biofilm formation on PUF after 28 days, indicating effective surface colonization and biomass retention. Toxicity assessment of treated samples from pilot-scale bioreactor showed substantial detoxification, with *Lemna minor* growth inhibition reducing from 98–100% in untreated effluent to 65–70% in treated samples. Likewise, seed germination assay showed an improvement in growth germination index (GGI%) from 3–5% (in untreated samples) to 75%, 58%, and 20% for treated samples using *Vigna radiata*, *Cicer arietinum*, and *Lactuca sativa* seeds, respectively, which reasonably confirmed the detoxification of treated effluent. Overall, the PUF integrated anaerobic bioreactor exhibited promising decolourization, COD removal, and toxicity reduction. However, incorporation of further aerobic polishing unit can remove residual COD and toxicity to a greater extent.

Keywords Textile effluent, Polyurethane (PUF), Bioremediation, Pilot-scale bioreactor, Dye degradation, Toxicity

Graphical abstract



1 Introduction

Modernisation and globalisation have driven the rapid expansion of textile industries over the past decade. Despite making significant contributions to job employment, growth, and development in a country, the textile industry releases dye-laden wastewater along with various organic (surfactants, sizing agents, phenols) and inorganic pollutants (heavy metals, synthetic dyes, sulfates, chlorides). Such effluents, if inadequately treated, can alter the physicochemical characteristics of receiving water bodies and can be toxic to biological entities [15]. Hence, a sustainable, cost-effective treatment strategy is therefore essential for mitigating textile-derived pollutants. Currently various physicochemical methods have been adopted by industries which include coagulation, filtration, adsorption and advanced oxidation processes like Ozonation, UV/Fenton, electrochemical oxidation have been adopted for the removal of dye wastewater [13, 38, 53]. However, high chemical and energy costs, sophisticated handling, generation of toxic sludge, and incomplete mineralization are some of the demerits of conventional processes [1, 43].

Biological treatment, including anaerobic and aerobic processes, uses microorganisms as an alternative method, as microbial cells can break down the persistent pollutants into simpler carbon sources [36, 55]. In aerobic systems like trickling filters, activated sludge and aerated lagoon systems these are generally used to reduce chemcial oxygen demand (COD), colour, biological oxygen demand (BOD), and other contaminants [3]. However, these systems are incapable of degrading dyes and also generate huge amount of sludge which is a major drawback [18, 29].

Conversely, anaerobic treatment has been recognized as a promising alternative for managing concentrated and highly recalcitrant dye effluents [60]. Several studies have demonstrated that anaerobic systems often outperform aerobic processes when treating high-strength wastewater streams [44, 52]. Extensive research has explored the biodegradation of dye-containing wastewater through various anaerobic reactor systems, including sequencing batch [56], fluidized bed [4], up flow anaerobic sludge blanket [20], rotating biological contactors [35], and attached or packed bed reactors [39]. These studies consistently demonstrated high levels of colour and pollutant removal efficiency [62].

However, owing to the inherent toxicity of textile effluents, many previous studies relied on extensive dilution with synthetic media and required costly additives or pre-treatment steps such as pH neutralization and cooling [57]. Consequently, the use of robust dye-degrading microorganisms immobilized on suitable attachment media is considered more effective and operationally stable than suspended systems for treating complex textile effluent.

Immobilized cells exhibit enhanced stability and are less susceptible to cell death under extreme environmental conditions such as variations in pH, temperature, and pollutant concentration. Furthermore, immobilization minimizes biomass washout and enhances substrate utilization efficiency, making it a preferred strategy for biological treatment processes [37, 46]. Various biocarriers, including polyurethane foam [6], luffa sponge [24], activated carbon [40], and polyethene [34], have been utilized in bioreactors for textile effluent treatment. Among these, polyurethane foam (PUF) has gained particular attention due to its non-biodegradable nature, high surface area with active sites favorable for microbial colonisation, excellent thermal stability, mechanical strength, and chemical inertness [23]. Recent studies have demonstrated its effectiveness in dye degradation systems; for instance, [53] reported 64.6–88.4% decolourization of Congo Red dye using *Lysinibacillus* immobilized on polypropylene-polyurethane foam (PP-PUF) in a moving bed biofilm reactor, while [54] achieved 87.31% removal of Acid Orange dye (300 mg/L) in a fixed-bed bioreactor packed with immobilized PUF.

However, these studies were often limited to the degradation of synthetic dye solutions under controlled laboratory conditions, which may not accurately represent the complex and variable composition of real textile effluents. Moreover, the toxic interactions and inhibitory effects of actual dye bath chemicals, surfactants, and salts on microbial activity have not been adequately addressed in prior studies. Consequently, polyurethane foam (PUF) based immobilization systems have shown promising decolorization in synthetic media. However, their scalability, operational stability, and long-term performance under real wastewater conditions remain insufficiently evaluated. Hence, in view of this the objectives of the present study were: 1) Development of a lab-scale bioreactor for the treatment of synthetic dye and textile effluent, 2) Scaling-up of the bioreactor to the pilot-scale for treatment of textile effluent from different industries, 3) Toxicity assessment of the treated effluent from the pilot-scale bioreactor using *Lemna minor* (duckweed plant), green bean (*Vigna radiata*), chickpea (*Cicer arietinum*), and lettuce (*Lactuca sativa*).

2 Materials and methods

2.1 Materials

Polyurethane foam (Commercial grade) and acrylic sheet used in this study for the bioreactor fabrication and operation was procured from the local market of New Delhi, India. Reactive blue 13 (RB13) dye powder was procured from a major textile manufacturing facility located in Madhya Pradesh, India. Yeast extract (Himedia: Cat. No. RM027) was used as nutrient source in all experiments.

2.2 Collection and procurement of textile dye effluent

The effluent samples were obtained from the same industrial facility to maintain consistency in pollutant origin and composition. The effluent was sourced from two operational units within the industry: the dyeing unit and the pre-treatment range (PTR) unit. The dyeing unit is responsible for the discharge of approximately 4.0—5.0 million liters per day (MLD) of effluent, whereas the PTR unit contributes an additional 1.2—1.6 MLD of effluent. Additional effluent samples, designated as Effluent-1 and Effluent-2, were collected from a textile industry located in Ghaziabad, India. All collected effluent samples were transported to the laboratory, where they were immediately characterized to quantify key physicochemical parameters including pH, electrical conductivity (EC), total dissolved solids (TDS), chemical oxygen demand (COD), and biological oxygen demand (BOD) following the standard protocols described in Analytical Sect. 2.5. Furthermore, the samples were stored at 4 °C to prevent any changes in composition before further use. The detailed characterization of all the procured samples is given in Table 1.

2.3 Optimization and decolourization experiments performed using polyurethane foam as a support matrix for the microbial consortium

The optimization of different packing rates (PR) of PUF on anaerobic dye decolourization and turbidity removal from treated effluent was done. The PUF cubes were cut to dimensions of 2.0 × 2.0 × 2.0 cm and used in various quantities to obtain three packing rates: 8% w/v (10 cubes), 16% w/v (20 cubes), 24% w/v (30 cubes). Afterwards, all cubes were washed thoroughly with milli-Q water and sterilized using an autoclave before use. In batch experiments, PUF was used in various packing rates as mentioned above. A reagent bottle without any packaging of PUF served as a control. Further, Reactive blue 13 dye (100 ppm): 70% v/v, PTR effluent: 30% v/v and 0.5 g/L yeast extract were fed into reagent bottles. Furthermore, inoculum (10% v/v) of a microbial consortium (OD_{600} = 0.380 ± 0.025) was added. The microbial consortium used in the present study was previously developed in our lab which includes genera like *Enterococcus, Delftia, Dyadobacter, Clostridium, Bacteroides, Dysgonomonas, and Ochrobactrum* (Samuchiwal et al., 2021). The experiment was performed in four successive cycles, each lasting 72 h. Sampling was done at regular intervals (0, 24, 48, and 72 h) using a syringe and analyzed for parameters like pH, colour (Hazen), and turbidity (NTU) as per the standard protocols mentioned in analytical Sect. 2.5. After every cycle, the treated liquid was removed and replaced with a fresh dye effluent mixture while retaining the microbial consortium attached to the PUF.

Table 1 Physiochemical characterization of Textile effluents used in the study

Parameters	Effluent samples (Madhya pradesh industry)		Effluent samples (Ghaziabad industry)	
	Colour effluent (Dyeing unit)	Non-colour effluent (Pre-treatment unit)	Effluent-1	Effluent-2
pH	10.20 ± 0.45	11.7 ± 0.50	10.20 ± 0.45	9.8 ± 0.20
COD (mg/L)	1085 ± 10.67	2312 ± 10.14	2885 ± 10.67	3270 ± 10.23
Colour (Hazen)	2380 ± 15.45	963 ± 12.28	2380 ± 5.35	4876 ± 6.45
Sulphate (mg/L)	380 ± 2.56	150 ± 2.32	380 ± 2.67	560 ± 2.56
TDS (mg/L)	3560 ± 25.34	1009 ± 18.29	6360 ± 12.43	8360 ± 14.34
EC (mS/cm)	10.60 ± 0.80	2.1 ± 0.40	10.60 ± 0.80	13.60 ± 0.80
Alkalinity (mg/L)	900 ± 2.45	350 ± 1.32	900 ± 2.45	1012 ± 2.45
Turbidity (NTU)	17 ± 5	64 ± 10	35 ± 10.67	39 ± 10.67

2.4 Fabrication and operation of PUF integrated lab and pilot-scale bioreactor

2.4.1 Lab-scale bioreactor

A lab-scale anaerobic bioreactor was designed and fabricated. The bioreactor was cylindrical, having dimensions of 28 cm height and 15 cm diameter with total working volume of 4 L. Inlet and outlet port were placed at 24 cm and 2 cm height, respectively with gas purging port and nutrient/inoculum port located at the top of the bioreactor as shown in Supplementary file (SF) 1A.

Further, to enhance microbial attachment and facilitate biofilm development, 24 cubes of PUF measuring 3.5 × 3.5 cm × 3.5 cm were prepared and introduced into the bioreactor. The PUF cubes were cut to appropriate size, ensuring an optimized packing density of 24.6% (w/v) within the bioreactor. These attachment materials served as the primary support matrix for microbial colonization. Another bioreactor having same dimensions and volume as mentioned above was fabricated to serve as control bioreactor (without PUF). The bioreactor was connected to a peristaltic pump to maintain controlled influent flow (at flow rate: 0.93 mL/min) and to uniformly distribute the Reactive blue 13 dye solution by circulation.

2.4.2 Pilot-scale bioreactor

The pilot-scale anaerobic bioreactor was fabricated at the industrial premises. The cylindrical bioreactor was constructed with a diameter of 600 mm and a total height of 1500 mm. The total working volume of the bioreactor was 200 L. The bioreactor was divided functionally into two zones: the active treatment zone of 1000 mm in height, filled with polyurethane cubes (dimensions: 4 cm x 4 cm x 4 cm) with a packing density of 25% w/v to enhance microbial retention, and a bottom settling zone of 500 mm for sludge accumulation and solid separation. Throughout the bioreactor, PUF cubes (dimensions: 4 cm x 4 cm x 4 cm) were uniformly distributed to provide surface area for microbial attachment and to improve treatment efficiency by supporting anaerobic biofilm formation (SF-1B). Further, an inlet port to feed the bioreactor was made at the top surface of the bioreactor. An outlet provision was made in the side wall at the bottom of the bioreactor to collect the treated effluent from the bioreactor.

2.4.3 Repeated fed-batch and continuous operation of PUF integrated lab-scale bioreactor using blue dye CP and textile effluent

The repeated fed batch experiments (8 cycles) were performed using the PUF bioreactor. Before starting the experiment, the PUF cubes were arranged inside the bioreactor at a defined packing density (24.6% w/v) to maximize surface area. For the influent, a dye solution was prepared by mixing 70% v/v Reactive Blue 13 dye at a concentration of 100 ppm along with 30% v/v pre-treatment effluent (PTR) and filled inside the bioreactor. Additionally, a nutrient source yeast extract at 0.5 g/L concentration was added to support microbial activity. The bioreactors were sealed and purged with nitrogen to develop anaerobic conditions inside the bioreactor. A 10% (v/v) inoculum (OD600 nm = 0.290) was added using a syringe. The hydraulic retention time (HRT) was set to 72 h, and samples were drawn from sample outlet port at specific intervals at the start (0 h), and after 24, 48, and 72 h. These sampling points allowed continuous monitoring of how effectively the system could increase the dye decolourization efficiency and reduce turbidity. After 72 h, 80% of the contents of bioreactor were drained and the bioreactor was fed again with 70% v/v Reactive Blue 13 dye (100 ppm) along with 30% v/v PTR and yeast extract at 0.5 g/L. After repeated fed batch mode study, the bioreactor was operated in continuous mode. A peristaltic pump was employed to transfer the effluent-1 (70% v/v dye effluent: 30% v/v PTR effluent) from the collection tank into the bioreactor using a silicon pipe (Inlet diameter: 6.4 mm). The pump was operated at a flow rate of 0.93 mL/min to transfer textile effluent to the inlet and to collect the treated effluent from the outlet, respectively.

2.4.4 Repeated fed-batch and continuous operation of PUF integrated onsite pilot-scale bioreactor for industrial textile effluents treatment

The pilot-scale anaerobic bioreactor was divided functionally into two zones: the active treatment zone filled with polyurethane cubes (dimensions: 4 cm x 4 cm) with a packing density of 25% w/v (previously optimized in lab-scale bioreactor) to enhance microbial retention, and a bottom settling zone for sludge accumulation and solid separation. Throughout the bioreactor, PUF cubes were uniformly distributed to provide surface area for microbial attachment and to improve treatment efficiency by supporting anaerobic biofilm formation. Further, an inlet port to feed the bioreactor was made at the top surface of the bioreactor. An outlet provision was made in the side wall at the bottom of the bioreactor to collect the treated effluent from the bioreactor. The effluent-2 was initially stored in a 300 L collection tank and was supplied into the bioreactor using a rubber pipe (diameter: 15 mm) through inlet port using a water pump (V-guard 0.25 hp single-phase self-priming pump) at a controlled flow rate of 2.778 L/hr. Furthermore, the PUF system was operated in continuous mode to evaluate its long-term performance under varying effluent conditions. The PUF anaerobic bioreactor was continuously fed with a 100% v/v textile along with yeast extract supplementation after every 72 h of operation. The effluent inlet and outlet flow rate was maintained at 2.78 L/hr. Samples were withdrawn at 0, 24, 48, and 72 h during the batch cycle and analyzed for pH, COD, turbidity (NTU), and colour (Hazen) as per the standard protocols mentioned in analytical Sect. 2.5

2.4.5 SEM analysis of PUF obtained from pilot-scale bioreactor before and after microbial attachment

Scanning electron microscopy (SEM) is a widely employed technique for visualizing bacterial attachment on material surfaces and assessing associated changes in surface morphology. To evaluate biofilm-induced alterations on the PUF matrix, samples were collected at defined time points (day 0 and day 28). SEM analysis was performed following the protocol described by Prajapati et al. (2015). In brief, PUF pieces containing attached microbial cells and debris were sectioned into small fragments, rinsed with 0.15 M phosphate-buffered saline (PBS, pH 7.2), and fixed in 2.5% glutaraldehyde at 4 °C overnight. Post-fixation, the samples were washed repeatedly with PBS and lyophilized to achieve complete drying. The dried specimens were then mounted on aluminum stubs, gold-coated by cathodic sputtering, and examined using a ZEISS EVO 50 scanning electron microscope at the Central Research Facility (CRF), IIT Delhi.

2.5 Toxicity assessment of treated samples from pilot-scale bioreactors

2.5.1 Phytotoxicity assessment using Lemna minor

Lemna minor (duckweed) was procured from a local market shop in New Delhi, India. Prior to experimentation, the plants were acclimatized and maintained in Steinberg medium for approximately two weeks. Acclimatization was conducted in a plastic tray with a volume of approximately 4 L, filled with 2 L of Steinberg medium. Lemna minor was cultured under controlled environmental conditions with a 16:8 h light–dark cycle using cool fluorescent illumination at 90 µmol m^{-2} s^{-1} and maintained at 24 ± 2 °C. The culture medium was adjusted to pH 7.4 to ensure neutral conditions and reduce physiological stress during acclimation, despite the species' documented ability to tolerate pH values ranging from 3.5 to 10.

Following the acclimation period, a toxicity assessment was conducted using three different wastewater samples: 1) Control (Tap water), 2) Raw textile effluent, 3) Anaerobically treated effluent. Each effluent sample was used at 100% concentrations (v/v) in plastic Petri plates, using Steinberg medium with a final volume of 10 mL. For each treatment, four healthy, freshly grown *L. minor* colonies (each with 6 fronds) were carefully selected from the acclimation tray and transferred into individual Petri plates. All treatments were conducted in triplicate to ensure statistical robustness and reproducibility.

The plates were incubated under the same controlled conditions described earlier for a period of 8 days. Growth and physiological responses of *L. minor* were monitored to evaluate the toxic effects of the effluents. Growth was assessed by measuring frond number, fresh weight (FW), and dry weight (DW) following the ISO 20079 standard protocol (2004). Frond counts were recorded on Day 0 and again on Day 6. Only visibly healthy fronds were included in the counts.

Fresh biomass was quantified by gently blotting the plants with paper towels and recording their weight using an analytical balance. For dry biomass determination, the plants were oven-dried at 80 °C for approximately 24 h until a constant weight was obtained. The relative growth rate (RGR) was then calculated using the following equation:

$$RGR = \frac{ln(N_t) - ln(N_0)}{t} (1)$$

where, N_t: Frond number at time 't' (end of the test), N_0: Initial frond number, t: Test duration in days.

Dry to fresh weight ratio (DW/FW) was determined according to equation no. (2)

$$\frac{DW}{FW} = \frac{Dryweight(g)}{Freshweight(g)} (2)$$

Percent growth inhibition was calculated from the formula:

$$\%inhibition = \frac{RGR_{control} - RGR_{sample}}{RGR_{control}} X100$$

2.5.2 Seed germination inhibition test using green bean (Vigna radiata), Chickpea (Cicer arietinum), and Lettuce (Lactuca sativa)

Phytotoxicity assessment of the test samples of Untreated textile effluent, anaerobically treated Effluent and Tap water was done using seed germination inhibition assay using Lettuce (*Lactuca sativa*), Green Bean (*Vigna radiata*), and Chickpea (*Cicer arietinum*) as it is sensitive to phytotoxic compounds in the test solutions (Mendes et al., 2021). In brief, 10 mL of the test solution was dispensed into a Petri dish lined with Whatman No. 1 ashless filter paper, after which ten seeds were placed in each dish, with three replicates per treatment. The plates were then incubated in the dark at 25 °C, and tap water served as the control. After six days, seed germination was recorded manually, and the germination index (GI%) was calculated using the following equation:

$$RSG\,(\%) = \frac{No.of seedsgerminatedinTestsolution}{numberof seedsgerminatedincontrol} X100$$

$$RRG\,(\%) = \frac{MeanrootlengthinTestSolution}{Meanrootlengthincontrol} X100$$

$$GI\,(\%) = RSG\% X RRG\%$$

where, RSG: relative seed germination; RRG: relative root growth; GI: germination index.

2.6 Analytical methods

The pH of the samples was measured using a benchtop pH meter (Eutech CyberScan PC510), while turbidity was quantified with a turbidity meter (Thermo Scientific TN-100). For COD and colour analyses, samples were centrifuged at 9200 g for 10 min using a microcentrifuge (Spinwin MC-01, Tarsons), and the resulting supernatant was used for further assessment. COD was determined following the APHA open-reflux method (5220B), wherein samples were digested with potassium dichromate ($K_2Cr_2O_7$) and subsequently titrated with ferrous ammonium sulfate. Reactive Blue 13 concentration was measured at 600 nm using a microplate reader (BioTek Eon). Effluent colour (Hazen units) was quantified using a Hazen meter (Lovibond MD100).

2.7 Statistical analysis

All data were statistically analyzed using Microsoft excel. Experiments were conducted in triplicate, and the resulting values are reported as the mean ± standard deviation (SD).

3 Result and discussion

3.1 Decolourization experiments employing polyurethane foam as support matrix for microbial consortium

Across four consecutive treatment cycles (1st – 4th), we assessed the impact of three different packing rates of PUF (8%, 16%, and 24% w/v) on anaerobic decolourization of blue dye CP and turbidity removal. After 72 h, 24% PUF packing rate achieved the highest decolourization at 85.2 ± 9.5%, significantly outperforming both 8% and 16% packing rates (which reached approximately 68.5 ± 6% 71.6 ± 6.9%, respectively) as shown in Fig. 1(A). Compared to control i.e. without PUF (treated dye turbidity, 70 NTU),

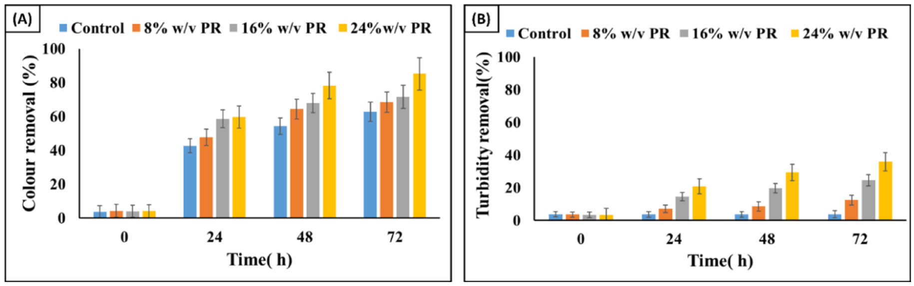


Fig. 1 Optimization of packing density in the PUF foam-integrated anaerobic bioreactor. **A** Colour removal efficiency at different PUF packing densities. **B** Turbidity removal efficiency at varying PUF packing densities

turbidity removal (Fig. 1(B) was at the highest ~ 73% at 24% w/v PUF (18.7 NTU) followed by 60% (27.36 NTU) at 8% w/v and 69.2% (21.5 NTU; 69.2%) at 16% w/v packing rates.

The performance of the 24% PUF packing rate can be attributed to an optimal balance between attachment of microbial cells and biomass transfer efficiency. At this higher packing rate, the PUF provides the maximum surface area for microbial attachment [23, 48] while still upholding appropriate pore space for dye access to microbial communities and nutrient diffusion [45],[31].

The progressive increase in decolourization and turbidity removal with increasing PUF packing rate highlights its role in enhancing microbial attachment [61]. Interconnected foam pores facilitate the colonization of microbial cells responsible for anaerobic dye degradation and promote the attachment of suspended matter through extracellular polymeric substances (EPS). Packing rate levels below 24% performed less, likely due to insufficient surface area for biomass accumulation and weaker attachment. Furthermore, at 24% w/v packing rate, PUF were densely packed into the bioreactor which resulted in overcrowding inside the bioreactor and were anticipated to pose considerable operational difficulties. Hence, experiments at higher densities were not attempted. Moreover, high packing rate could restrict mass transfer [23]. Optimal packing rates and material selection therefore balance increased surface area for attachment with adequate mass transfer to support microbial activity [27]. Our findings signify that 24% packing rate gave perfect stability to support anaerobic bioreactor performance.

3.2 Performance evaluation of PUF integrated lab and pilot-scale bioreactors

3.2.1 Lab-scale bioreactor

3.2.1.1 Repeated fed batch mode study of PUF bioreactor with blue dye CP dye The study observed the decolourization of Reactive Blue 13 dye (100 ppm) in repeated fed batch mode setup using a 4L prototype anaerobic bioreactor. The data shown here is the average of Cycle 3 to cycle 6 as the first 2 cycles were not considered for data collection considering the initial stabilization period. The bioreactor with PUF as the attachment media demonstrated a decolourization efficiency of 86.2 ± 1.15% after 72 h. In comparison, the decolourization efficiency of the control bioreactor (without PUF) showed a lower decolourization efficiency at ~ 67.8 ± 1.10% (Fig. 2 A).

The significant colour reduction as compared to positive control highlights that the presence of PUF likely contributed to enhanced efficiency by providing a more porous structure for microbial growth and attachment which leads to enhanced degradation of the dye [48]. After 72 h, the turbidity removal efficiency in the PUF bioreactor reached

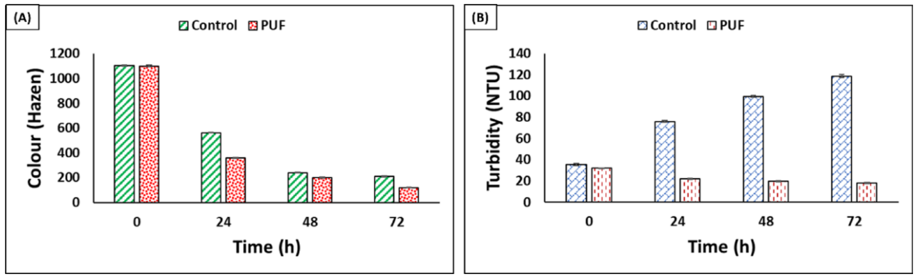


Fig. 2 Performance assessment of PUF integrated anaerobic reactor for reactive blue 13 dye (100 mg/L): **A** Colour removal 80.9 ± 1.7% (average of 3 cycle) **B** Turbidity removal by PUF 43.74 ± 1.8% (average of 3 cycles) at batch mode operation after 72 h of incubation time

approximately 80.9 ± 1.7% compared to the control. The final turbidity at the outlet of the PUF bioreactor was 18.5 ± 1.8 NTU, whereas the control exhibited a significantly higher turbidity of 97.0 ± 2.3 NTU (Fig. 2 B). The presence of PUF seems to have two benefits, first supporting microbial activity [16] and second helping in the physical removal of suspended solids which results in the less turbid effluent. Stable microbial activity under experimental circumstances in anaerobic bioreactor with PUF was also demonstrated by continuously maintained pH at 7.0. Moreover, no significant changes in pH were observed. Hence, stable pH is essential for consistent microbial activity in anaerobic bioreactors [7] which eventually affect the decolourization efficiency [11]. Furthermore, the weight of attachment material was observed at different intervals of days. After 30 days it was observed that the weight of the attachment material (PUF) increased by around 27.3%.

This weight gain supports the theory that the accumulation of microbial biomass on the PUF surface and in the pores of PUF serves as an effective material for microbial attachment [28]. The increase of foam weight suggests that it can withstand microbial populations for extended periods of time and improve the long-term treatment performance [59]. The results from the anaerobic bioreactor with PUF showed that the addition of attachment materials significantly enhances the efficiency of microbial consortia for coloured effluent treatment. The decolourization and turbidity removal efficiencies observed in the PUF bioreactor indicate that it provides a favorable micro-environment for microbial activity. This enhanced microbial performance consequently improves suspended solids and colour removal from the treated effluent.decolourization. The findings from this study provide strong evidence favoring the use of PUF as an attachment material in anaerobic bioreactors for efficient textile effluent treatment. The significant decolourization and turbidity removal efficiencies with the stable pH underline need for its pilot-scale trials in industrial premises under continuous mode.

3.2.1.2 Continuous mode study on textile effluent After repeated fed batch mode study, the bioreactor was operated in continuous mode. The inlet and outlet flow rate were maintained at a flow rate of 0.93 mL/min, and the results are shown in (Fig. 3). The colour (Hazen) removal of ~ 63.14 ± 10% was achieved in PUF bioreactor as compared to a value of 49.37 ± 9.5% for control bioreactor. This substantial reduction of colour can be attributed to the increased surface area and porosity of the PUF, which promoted microbial density inside the PUF bioreactor. Further, the turbidityan indicator of suspended and colloidal particles in the effluent, achieved a reduction to 18.35 ± 1.7 NTU in PUF biore-

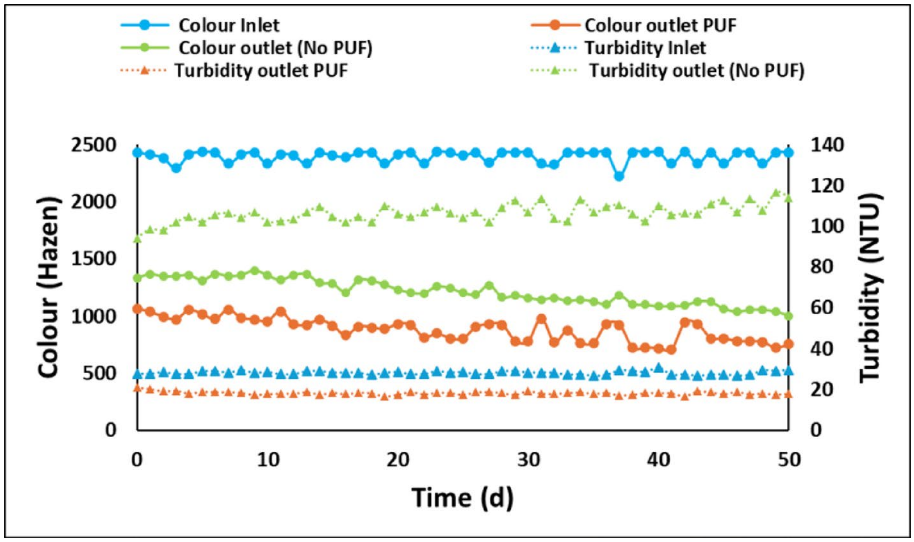


Fig. 3 Continuous operation of lab-scale PUF integrated anaerobic bioreactor showing colour (Hazen) and Turbidity (NTU) values in comparison to control bioreactor (No PUF)

actor outlet compared to turbidity control outlet (106.36 ± 1.4 NTU). Moreover, the PUF containing bioreactor maintained a stable average pH of 7.5 ± 0.57.

These findings align with previous study emphasizing the role of biofilm-based systems in improving dye removal efficiency under continuous mode conditions [32]. The COD was reduced by 60.3 ± 8.6% in the outlet of treated effluent as compared 40.17% ± 10.6% in control bioreactor. The enhanced COD removal is indicative of fast degradation of recalcitrant organics in PUF based anaerobic bioreactor [50], Lamssali et. al., 2024). The PUF may have supported higher biomass retention leading to prolonged contact time between the pollutants and active microbial populations [25]. These observations proved the suitability of PUF -based biofilm bioreactors for effective treatment of coloured textile effluent.

Furthermore, the improvement suggests that the physical entrapment of solids within the porous structure of the foam played a significant role in effluent turbidity removal. Another study also showed that PUF supports long-term continuous bioreactor operation with sustained high removal efficiencies in terms of colour and turbidity even under varying dye concentrations and flow rates [58].

3.2.2 Pilot-scale bioreactor

3.2.2.1 Repeated fed-batch operation of pilot-scale PUF bioreactor During a batch-mode study in a 200 L PUF anaerobic bioreactor, the performance of the system was evaluated across a 72-h cycle, and data represented here are average values from the 3rd to the 6th operational cycles, confirming the bioreactor's stability and reproducibility. Within the first 24 h, approximately 55 ± 4.8% color removal was achieved along with turbidity reduction, suggesting that microbial biomass had begun to colonize on the surface as well as inside the pores of PUF. The pH decreased from a highly alkaline level of ~ 11.5 to 9.8, indicating the onset of microbial activity and partial buffering within the system. Moreover, COD removal at this stage was around 35 ± 2.5%, demonstrating that initial degradation of organic pollutants had begun, possibly through hydrolysis and fermentation pathways. By 48 h, the bioreactor performance had improved considerably. Color removal rose to approximately 70 ± 5.7% and turbidity removal further to 66.02 ± 3.9% (from 102.66 to 102.33 NTU), highlighting ongoing microbial degradation of dye compounds and attachment of microbial biomass. The pH continued to decline, reaching

8.5 ± 0.28. Moreover, there was a substantial increase in COD removal, which reached 60 ± 2.5%, suggesting that the microbial community efficiently degraded complex organic matters. At the end of the 72-h cycle, the system exhibited optimal performance. Color removal peaked at 76.86 ± 5.7%, indicating that most of the dye compounds had been broken down by the microbial consortia. Turbidity reduction reached 30.96 ± 4.6% (from 10,333 to 72 NTU), reflecting further reduction in the treated effluent by PUF anaerobic bioreactor. Notably, the pH stabilized at 7.06, a near-neutral condition ideal for anaerobic microbial activity. COD removal efficiency also reached a maximum of 76.14 ± 5.6%, confirming that a significant portion of the organic load had been mineralized. This stabilization indicates a balanced metabolic state within the bioreactor, where fatty acid production has significantly lowered pH levels toward neutrality [63]. Further, the effectiveness of microbial colonization on the PUF attachment material was evaluated through dry weight measurement over 30 days. A consistent 22–25% increase in attachment material weight was recorded, providing direct evidence of sustained biofilm formation. This biofilm not only contributed to the degradation of pollutants but also offered structural flexibility to the system, enhancing process stability under shock loadings [34].

3.2.2.2 Long term performance assessment in continuous mode The PUF bioreactor was operated in continuous mode for 71 days in industrial premises under ambient conditions (Fig. 4A and B). The untreated effluent showed that colour remained relatively consistent throughout the study, with values ranging from 2600 ± 10.6 to 3200 ± 11.2 (Hazen). Despite the persistently high effluent load, the system maintained a steady decline in outlet color, stabilizing between 800 ± 10.3 and 1000 ± 9.2 (Hazen). A maximum decolourization efficiency of 76 ± 6.7% was achieved and sustained throughout stable-state operation. This consistent performance highlights the flexibility of the microbial community that was developed on the PUF. The dense and stable biofilm played a pivotal role in dye breakdown, likely through reductive cleavage of azo bonds under anaerobic conditions. The ability of the bioreactor to sustain such high decolourization efficiency under continuous flow reaffirms its scalability and field scale applicability, especially in decentralized industrial setups. Turbidity, which ranged between 40 ± 2.5 to 45 ± 2.9 NTU in the effluent, was another important parameter used to evaluate treatment performance. During the first 37 days, the system achieved an average turbidity reduction of ~ 38% was observed, with outlet turbidity levels ranging between 20 to 25 NTU. This reduction can be attributed to microbial flocculation and attachment of microbial biomass facilitated by the developed anaerobic biofilm. However, the turbidity removal plateaued after this point, indicating a need for further removal. To address this, a 1-micron filtration unit (ASP-20-HV1) was integrated on day 51, which dramatically improved effluent clarity. Post-filtration, turbidity levels dropped to 13 to 15 NTU, and overall removal efficiency increased. This two-

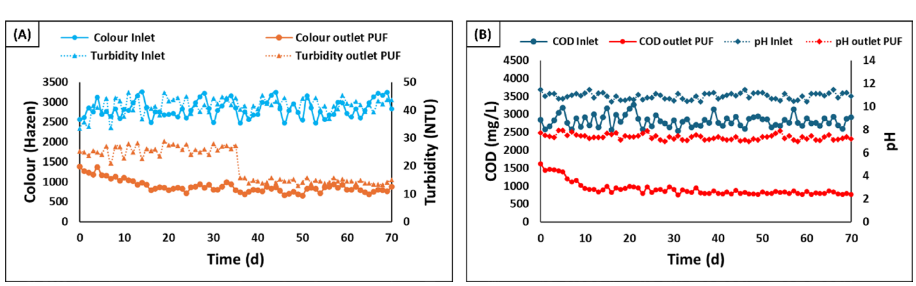


Fig. 4 Performance of the pilot-scale PUF-integrated anaerobic bioreactor under continuous operation at industrial conditions. **A** Colour (Hazen) and turbidity (NTU) variations in the outlet stream. **B** Chemical oxygen demand (COD) and pH variations in the outlet stream

stage setup of anaerobic reactor followed by membrane filtration proved essential for removing fine colloidal particles, thereby enabling the system to meet effluent discharge standards. Importantly, the PUF-supported biofilm ensured baseline stability, while the filtration unit provided the final polishing step for enhanced clarity in treated effluent. The system's capacity to remove organic load was assessed through COD measurements over the 71 day period. The inlet COD remained consistently high, ranging between 2600 and 3100 mg/L, reflecting the complexity and strength of textile effluent. The outlet COD, however, stabilized between 800 and 1000 mg/L after an initial acclimatization phase. A maximum COD removal efficiency of 71.97 ± 4.5% was achieved and remained stable beyond day 25, indicating that the bioreactor had reached a stable metabolic state.

The successful colonization of microbes on the PUF attachment material played a central role in this outcome, enhancing biomass retention, improving substrate-microbe contact, and preventing sludge washout. The anaerobic biofilm developed on PUF effectively mineralized complex organic matter demonstrating the bioreactor's ability to handle continuous organic loading without performance loss. Additionally, one of the key challenges in treating textile wastewater is managing the extremely alkaline pH of the influent, which in this case ranged from 10.5 to 11.5. Despite this, the bioreactor demonstrated remarkable buffering capacity, with the outlet pH stabilizing within the range of 7.1 to 8.0 throughout the study period. The lowest recorded pH was 7.1, is a crucial prerequisite for compliance with aquatic discharge standards and for minimizing soil alkalinity hazards during irrigation reuse.. Additionally, the PUF likely provided localized microenvironments within the biofilm that protected microbial populations from pH fluctuations, further stabilizing system performance [2]. The synchronized performance across all treatment parameters colour, turbidity, COD, and pH demonstrate the robustness and adaptability of the PUF-supported anaerobic bioreactor under long-term continuous operation. Although the results represent substantial pollutant abatement, direct discharge or unrestricted irrigation reuse still requires further reduction in COD, colour, and residual toxicity. This requirement is particularly stringent in regions enforcing zero-liquid-discharge (ZLD) policies or strict surface water discharge norms for textile effluents.

3.2.3 SEM analysis of PUF from pilot-scale bioreactor

The SEM images revealed structural changes in the porous structure over a period of 30 days indicating a clear tendency of decreasing pore size attributed to progressive biofilm formation. On Day 0, the structure was clean and untarnished, large well-defined pores were clearly visible on the surface, representing a pristine state with no microbial attachment.

As attachment of microbial biomass on the PUF begins, the surface changes start to emerge from the 7th day. The pores were still largely integral but a weak obscure around the edges indicated the early stages of microbial attachment. Bacterial attachment to surfaces is influenced by environmental, bacterial, and surface properties, leading to the formation of biofilms [10]. These signs became more noticeable after the 14th day. The visible pores were no longer open,their walls started to thicken, suggesting that microbes have begun secreting extracellular polymeric substances (EPS), gradually masking the original architecture. Several studies have reported that secretion of EPS is an essential microbial adaptation, enabling microbes to be attached to surfaces forming biofilms, and create organized microbial communities [26]. Further, other studies have also reported that micro and nanoscale features can either promote or improve bacterial attachment depending on their size, shape, and distribution [14, 21]. The real changes appeared on 21st Day. Under higher magnification (600x), the porous surface looks significantly changed. Biofilm layers began to develop, and the open pores were colonized by microbial biomass. The transformation was completed by the 28th day. SEM Images at 0 days and after 28th day have been shown in (Fig. 5A and B). The biofilm appears firmly attached, and it spreads deep into the pores, clearly signifying the attachment of biomass over time. As biofilms mature, they thicken and further reduce pore size (Jafari et al., 2018). Increased biofilm on PUF increases hydraulic resistance, which protects microbial biomass against shock loadings (Carpio et al., 2019).

3.3 Toxicity assessment of treated samples from pilot-scale bioreactor

3.3.1 Phytotoxicity assessment using Lemna minor

The toxicity assessment was conducted using *Lemna minor* (duckweed) on textile effluent, PUF-based anaerobically treated effluent, and Tap water samples (Fig. 6A). To evaluate the residual phytotoxicity of anaerobically treated textile effluent, a *Lemna minor* bioassay was conducted over a 6-day period. The experiment was designed to assess the relative inhibition of biomass growth in comparison to untreated effluent and control (tap water). Visual observations were supported by quantitative inhibition measurements to understand the impact of effluent toxicity on aquatic plant health. On Day 0, all test groups, including tap water (Control), untreated textile effluent, and anaerobically treated effluent, showed comparable healthy green fronds of *Lemna minor*. Marked differences in plant health became evident by day 6. The control group (tap water) maintained healthy and green fronds with no visible signs of damage, reflecting optimal growth conditions. In contrast, the samples exposed to untreated textile effluent exhibited significant chlorosis, frond disintegration, and necrosis, with nearly all biomass showing signs of decay. The calculated biomass inhibition ranged between 98–100%, indicating severe toxicity associated with the high concentration of dyes, salts, and residual chemical oxidants present in raw textile wastewater [42];[30];[22]. Interestingly, plants exposed to the anaerobically treated textile effluent displayed moderate signs of stress, including partial chlorosis and some necrosis, but fronds were still visibly intact and greener compared to the untreated group. Furthermore, the corresponding biomass inhibition was reduced to approximately 65–70%, suggesting that anaerobic treatment

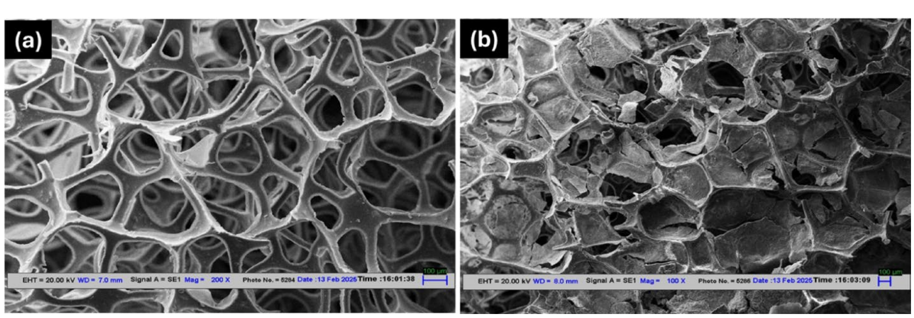


Fig. 5 Scanning electron micrographs (SEM) depicting the surface morphology of polyurethane (PUF) foam before and after biofilm formation. **A** Pristine PUF showing a smooth, porous structure. **B** PUF after reactor operation exhibiting dense microbial colonization and extracellular matrix deposition, confirming effective biofilm attachment

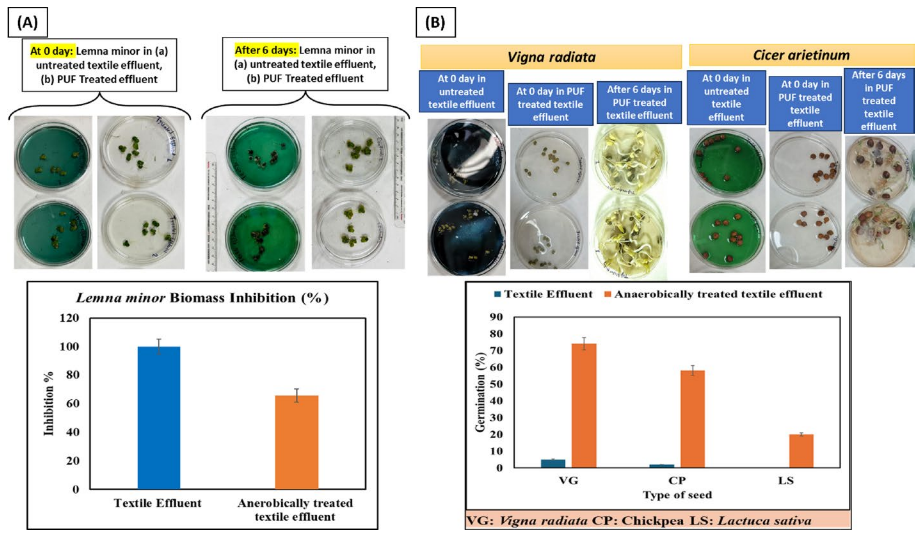


Fig. 6 Toxicity assessment of untreated and PUF-treated textile effluents using plant-based bioassays. **A** *Lemna minor* growth and biomass inhibition (%) after exposure to untreated and pilot-scale PUF integrated bioreactor treated samples for 6 days. **B** Seed germination response of *Vigna radiata* (mung bean), *Cicer arietinum* (chickpea), and *Lactuca sativa* (lettuce) showing reduced toxicity and improved germination after anaerobic-PUF treatment

significantly detoxified the effluent, yet residual inhibitory substances persisted [47], [12]. This partial toxicity can be attributed to incomplete mineralization of dye intermediates or the presence of aromatic amines produced after anaerobic dye degradation [42]. These findings suggest that phytotoxicity following anaerobic treatment was reduced but not completely eliminated. Therefore, there is a requirement of an additional post-treatment steps, such as aeration or advanced oxidation, to meet environmental discharge or reuse standards. The recommendation for post-aeration is based on the hypothesis that residual color and intermediate by-products could be further oxidized or volatilized, thereby improving the effluent's ecological compatibility. The observed phytotoxicity in anaerobically treated and raw textile effluent is likely attributable to pollutants such as aromatic amines and heavy metals, respectively, which can induce oxidative stress, disrupt physiological processes, and compromise cellular integrity in *L. minor* [9]; Žaltauskaitė et al., 2014). Previous studies have similarly reported up to 60% growth inhibition in duckweed exposed to untreated textile effluents [19]. In conclusion, *L. minor* proved to be effective bioindicators for evaluating effluent toxicity and treatment performance, offering critical insights for environmental risk assessment and the development of sustainable wastewater management strategies [41]; [8].

3.3.2 Seed germination inhibition test using green bean (Vigna radiata), Chickpea (Cicer arietinum), and Lettuce (Lactuca sativa)

The seed germination inhibition assay was performed using lettuce seeds (*L. sativa*), Green Bean (*Vigna radiata*) Chickpea (*Cicer arietinum*), pertaining to their sensitivity to phytotoxic compounds [51]. The results observed show that the highest seed germination percentage (90%) was associated with the control group having only tap water without any pollutants. To assess the residual phytotoxicity of anaerobically treated textile effluent on agricultural crops, a seed germination inhibition test was conducted using three plant species with differing sensitivity (Fig. 6 B): a) *Vigna radiata* (VG), b) *Cicer arietinum* (CP), and c) *Lactuca sativa* (LS). The germination percentage of seeds exposed to untreated and anaerobically treated effluents was recorded and compared to evaluate the impact of treatment on effluent toxicity. As depicted in the germination profile, seeds exposed to untreated textile effluent exhibited severely inhibited germination across all three species, with values ranging between 3–5%, indicating near-complete inhibition.

This reflects the high toxicity of untreated dye-laden wastewater, likely due to the presence of azo dyes, salts, heavy metals, and chemical additives used in textile processing. Upon anaerobic treatment, a notable improvement in seed germination was observed, although variability in sensitivity was evident among the species tested. *Vigna radiata* demonstrated the highest tolerance, showing a germination rate of approximately 75%, followed by Chickpea with ~ 58% germination. This suggests that anaerobic degradation significantly reduced phytotoxic constituents, enabling partial recovery of seed viability. The enhanced germination under treated effluent conditions can be attributed to the microbial breakdown of toxic dye compounds and a reduction in chemical oxygen demand, as observed in previous bioreactor performance data. In contrast, *Lactuca sativa* exhibited the lowest germination rate (~ 20%) even under anaerobically treated conditions, indicating higher sensitivity to residual metabolites or partially degraded dye intermediates. The differential response across plant species underscores the importance of crop-specific toxicity assessment when evaluating the reuse potential of treated effluents for irrigation. These results align with the findings of the *Lemna minor* bioassay, reinforcing the conclusion that while anaerobic treatment significantly reduces effluent toxicity, complete detoxification may not be achieved without additional post-treatment steps such as aeration or advanced oxidation. Also, the germination index (GI%) for Textile effluent was just 14%, which indicates severe phytotoxicity, as GI values below 50% typically suggest high levels of environmental stress on plant growth (Ravindran et al., 2016). A significantly higher toxicity was observed for the test solution of PUF based anaerobic treatment where the germination was completely inhibited (0%). The complete inhibition of germination and root elongation suggests that certain anaerobic metabolic intermediates like aromatic amines formed as a result of incomplete degradation of Azo dyes are potentially toxic to plant systems. For instance, a study by [49] demonstrated that the degradation of Acid Orange 7 by halophilic microbes resulted in the formation of carcinogenic and mutagenic aromatic by-products such as aniline and 1-amino-2-naphthol which was further shown to inhibit plant growth and caused chromosomal abnormalities in Allium cepa. Moreover, the study by [5] also demonstrated that the degradation products of azo dyes, especially the aromatic amines, exhibit higher toxicity levels that adversely affect the germination and growth rate of rice seeds. Collectively, these findings demonstrate that the PUF-integrated anaerobic bioreactor functions as a robust and industrially viable primary treatment unit, achieving substantial reductions in colour, COD, and toxicity. However, the results also indicate that anaerobic treatment alone is insufficient to consistently meet irrigation reuse or final discharge quality standards, necessitating a subsequent aerobic or advanced oxidation polishing step for complete mineralization and ecological safety. Furthermore, integration of the PUF bioreactor with appropriate post-treatment units can facilitates effective polishing, ensures regulatory compliance, and enables safe effluent reuse.

4 Conclusion

The present study demonstrated the successful development and scale-up of a PUF foam integrated anaerobic bioreactor for the treatment of textile dye effluents from different textile industries. Results from laboratory-scale bioreactor optimization indicated that a PUF packing density of 24% (w/v) provided the best performance, achieving 85.2 ± 9.5% color removal and ~ 73% turbidity reduction within 72 h. Scale-up to a 200 L pilot-scale system operated under real industrial conditions further validated the robustness of the process, with stable color and COD removal efficiencies of 76 ± 6.7% and 71.9 ± 4.5%, respectively, during 71 days of continuous operation. SEM analysis confirmed dense and uniform biofilm formation on the PUF matrix, which enhanced microbial retention, pollutant degradation along with overall bioreactor stability. Furthermore, time-resolved SEM highlights biofilm maturation on PUF (from Day 0 to Day 28), thereby demonstrating a link between micro-scale colonization on PUF and reactor-level stability during shocks. The dual toxicity bioassays (*Lemna minor* and seed germination) confirm substantial detoxification after anaerobic treatment, yet also reveal residual inhibition, underscoring the realistic need for a light polishing step (e.g., aerobic/oxidative) for complete toxicity removal. However, the partial inhibition persisted due to incomplete mineralization and formation of aromatic amine intermediates. Despite these promising results, certain limitations were identified. The anaerobic process alone was insufficient for complete detoxification, necessitating a subsequent aerobic or advanced oxidation step to achieve full mineralization. Additionally, operational period was limited to 71 days. Hence, long-term performance, biofilm aging, and maintenance requirements need further investigation. Overall, the PUF-integrated anaerobic bioreactor demonstrated effective performance and can be a suitable option for the primary treatment of textile effluents. Its integration with suitable post-treatment units could advance the development of hybrid treatment frameworks capable of achieving complete detoxification and regulatory compliance in industrial wastewater management.

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1007/s43621-025-02549-z.

Supplementary Material 1

Acknowledgements

The authors gratefully acknowledge partial financial support from Accelerated Translational Grant for Commercialization (ATGC), Department of Biotechnology, Government of India (BT/ATGC/127/SP39027/2020). Anushree Malik is grateful to Rita Duggal Chair Professorship at IIT Delhi (IITD/ADFA/2024/293403) for providing an encouraging ecosystem of industrial research.

Author contribution

Aman Chaudhary: Methodology, Investigation, Analysis, Conceptualization, Review & Editing, Writing original draft. **Nagesh Vikram Singh:** Conceptualization, Methodology, Investigation, Analysis, Writing original draft, Writing—Review & Editing. **Saurabh Samuchiwal:** Conceptualization, Writing—Review & Editing, Validation. **Vivek Kumar Nair:** Conceptualisation, Writing—Review & Editing **Bhupendra Singh Butola:** Fund acquisition, Supervision, Review & Editing, Validation, **Anushree Malik:** Fund acquisition, Supervision, Resources, Validation, Writing—review & editing.

Data availability

The datasets generated during and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethical approval and consent to participate

Not applicable.

Consent to publish
Not applicable.

Competing interests
The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Received: 10 November 2025 / Accepted: 22 December 2025

Published online: 08 January 2026

References:

1. Ahmad AL, Chin JY, Harun MHZM, Low SC. Environmental impacts and imperative technologies towards sustainable treatment of aquaculture wastewater: a review. J Water Process Eng. 2022;46:102553. https://doi.org/10.1016/j.jwpe.2021.102553.
2. Ahmad M, Liu S, Mahmood N, Mahmood A, Ali M, Zheng M, et al. Effects of porous carrier size on biofilm development, microbial distribution and nitrogen removal in microaerobic bioreactors. Bioresour Technol. 2017;234:360–9.
3. Ananthashankar R, Ghaly A. Photocatalytic decolourization of textile effluent containing reactive red 120 dye with UV/TiO2. Am J Engg Appl Sci. 2013;6(3):252.
4. Agrawal S, Tipre D, Dave SR. Biotreatment of azo dye containing textile industry effluent by a developed bacterial consortium immobilised on brick pieces in an indigenously designed packed bed biofilm reactor. World J Microbiol Biotechnol. 2023;39(3):83.
5. Anburaj J, Kuberan T, Sundaravadivelan C, Kumar P. Biodegradation of azo dye by *Listeria* sp. Int J Environ Sci. 2011;1(7):1760.
6. Arikan EB, Isik Z, Bouras HD, Dizge N. Investigation of immobilized filamentous fungi for treatment of real textile industry wastewater using up flow packed bed bioreactor. Bioresour Technol Rep. 2019;7:100197. https://doi.org/10.1016/j.biteb.2019.100197.
7. Atasoy M, Cetecioglu Z. The effects of pH on the production of volatile fatty acids and microbial dynamics in long-term reactor operation. J Environ Manage. 2022;319:115700.
8. Avinash GP, Namasivayam S, Prakash P, Shree SK. Effectiveness of pre-treated *Lemna minor* biomass in removing methyl orange dye from aqueous solutions: adsorption efficacy, kinetics, and ecotoxicity analysis. Biomass Convers Biorefin. 2025. https://doi.org/10.1007/s13399-025-06610-9.
9. Aslanzadeh M, Saboora A, Moradlou O. Phytoremediation potential of duckweed (*Lemna minor* L.) for hexavalent chromium removal in synthetic wastewater: unveiling physiological response and defense mechanisms against excessive heavy metal uptake. Int J Environ Sci Technol. 2024;21(16):10155–74.
10. Berne C, Ellison CK, Ducret A, Brun YV. Bacterial adhesion at the single-cell level. Nat Rev Microbiol. 2018;16(10):616–27.
11. Cao J, Sanganyado E, Liu W, Zhang W, Liu Y. Decolourization and detoxification of direct blue 2B by indigenous bacterial consortium. J Environ Manage. 2019;242:229–37. https://doi.org/10.1016/j.jenvman.2019.04.067.
12. Chaturvedi A, Rai BN, Singh RS, Jaiswal RP. Comparative toxicity assessment using plant and luminescent bacterial assays after anaerobic treatments of dyeing wastewater in a recirculating fixed bed bioreactor. J Environ Chem Eng. 2021;9(4):105466.
13. Cheng G, Wan J, Li Q, Sun L, Zhang Y, Li Z, et al. Degradation of reactive brilliant red X-3B by photo-Fenton-like process: effects of water chemistry factors and degradation mechanism. Water. 2022;14(3):380. https://doi.org/10.3390/w14030380.
14. Cheng Y, Feng G, Moraru CI. Micro- and nanotopography sensitive bacterial attachment mechanisms: a review. Front Microbiol. 2019;10:191.
15. Chung KT. Azo dyes and human health: a review. J Environ Sci Health C. 2016;34(4):233–61.
16. Dacewicz E. The application of biofiltration with polyurethane foams for domestic sewage treatment. J Ecol Eng. 2019;20(2):131–40. https://doi.org/10.12911/22998993/96342.
17. Dacewicz E, Lenart-Boroń A. Waste polyurethane foams as biomass carriers in the treatment process of domestic sewage with increased ammonium nitrogen content. Materials. 2023;16(2):619.
18. Dafale, N., Wate, S., Meshram, S., & Neti, N. R. (2010). Bioremediation of wastewater containing azo dyes through sequential anaerobic–aerobic bioreactor system and its biodiversity. *Environmental Reviews*, *18*(NA), 21–36.
19. de Alkimin GD, Paisio C, Agostini E, Nunes B. Phytoremediation processes of domestic and textile effluents: evaluation of the efficacy and toxicological effects in *Lemna minor* and *Daphnia magna*. Environ Sci Pollut Res Int. 2020;27(4):4423–41.
20. de Almeida R, de Souza Guimarães C. Up-flow anaerobic sludge blanket reactors in dye removal: mechanisms, influence factors, and performance. In: Biological Approaches in Dye-Containing Wastewater, vol. 1. Singapore: Springer; 2022. p. 201–27.
21. Elbourne A, Chapman J, Gelmi A, Cozzolino D, Crawford RJ, Truong VK. Bacterial-nanostructure interactions: the role of cell elasticity and adhesion forces. J Colloid Interface Sci. 2019;546:192–210.
22. Farid M, Sajjad A, Asam ZUZ, Zubair M, Rizwan M, Abbas M, et al. Phytoremediation of contaminated industrial wastewater by duckweed (*Lemna minor* L.): growth and physiological response under acetic acid application. Chemosphere. 2022;304:135262.
23. Feng Q, Wang Y, Wang T, Zheng H, Chu L, Zhang C, et al. Effects of packing rates of cubic-shaped polyurethane foam carriers on microbial community and removal of organics and nitrogen in moving bed biofilm reactors. Bioresour Technol. 2012;117:201–7. https://doi.org/10.1016/j.biortech.2012.04.076.
24. Fernandes CD, Nascimento VRS, Meneses DB, Vilar DS, Torres NH, Leite MS, et al. Fungal biosynthesis of lignin-modifying enzymes from pulp wash and Luffa cylindrica for azo dye RB5 biodecolourization. J Hazard Mater. 2020;399:123094.
25. Fia FR, Matos AT, Borges AC, Fia R, Cecon PR. Treatment of wastewater from coffee bean processing in anaerobic fixed bed reactors with different support materials. J Environ Manage. 2012;108:14–21. https://doi.org/10.1016/j.jenvman.2012.04.033.

26. Flemming HC, Baveye P, Neu TR, Stoodley P, Szewzyk U, Wingender J, et al. Who put the film in biofilm? NPJ Biofilms Microbiomes. 2021;7(1):10.
27. Ghofrani-Isfahani P, Tsapekos P, Peprah M, Kougias P, Zervas A, Zhu X, et al. Ex-situ biogas upgrading in thermophilic trickle bed reactors packed with micro-porous materials. Chemosphere. 2022;296:133987.
28. Goswami L, Pakshirajan K, Pugazhenthi G. Biological treatment of biomass gasification wastewater using *Rhodococcus opacus* in an up-flow packed bed bioreactor with a nano-biochar support. J Clean Prod. 2020;256:120253.
29. Holkar CR, Jadhav AJ, Pinjari DV, Mahamuni NM, Pandit AB. A critical review on textile wastewater treatments: possible approaches. J Environ Manage. 2016;182:351–66.
30. Ishaq HK, Farid M, Zubair M, Alharby HF, Asam ZUZ, Farid S, et al. Efficacy of *Lemna minor* and *Typha latifolia* for the treatment of textile industry wastewater in a constructed wetland under citric acid amendment: a lab scale study. Chemosphere. 2021;283:131107.
31. Kaul A, Böllmann A, Thema M, Kalb L, Stöckl R, Huber H, et al. Combining a robust thermophilic methanogen and packing material with high liquid hold-up to optimize biological methanation in trickle-bed reactors. Bioresour Technol. 2022;345:126524. https://doi.org/10.1016/j.biortech.2021.126524.
32. Li J, Gong JL, Zeng GM, Zhang P, Song B, Cao WC, et al. Zirconium-based metal-organic frameworks loaded on polyurethane foam for simultaneous removal of dyes. J Colloid Interface Sci. 2018;527:267–79.
33. Lamssali M, Mantripragada S, Deng D, Zhang L. Enhancing swine wastewater treatment: a sustainable and systematic approach through optimized chemical oxygen demand/sulfate mass ratio in attached-growth anaerobic bioreactor. Environments. 2024;11(8):162.
34. Mahto K, Kumari S, Das S. Unraveling the complex regulatory networks in biofilm formation and its relevance in remediation. Crit Rev Biochem Mol Biol. 2021;57:305–32.
35. Madhushika, H. G. (2021). Decolourization of textile dyes in a fixed bed biofilm reactor using native microorganisms (Doctoral dissertation, University of Moratuwa).
36. Maurya KL, Swain G, Sonwani RK, Verma A, Singh RS. Biodegradation of Congo red dye using polyurethane foam-based biocarrier with activated carbon and sodium alginate. Bioresour Technol. 2022;351:126999.
37. Mehrotra T, Dev S, Banerjee A, Chatterjee A, Singh R, Aggarwal S. Use of immobilized bacteria for environmental bioremediation: a review. J Environ Chem Eng. 2021;9(5):105920.
38. Modkovski TA, Maciel GM, Brugnari T, Bortolini DG, Bergamasco R, Treichel H, et al. Simultaneous removal of textile dyes by adsorption and biodegradation using *Trametes villosa* laccase immobilized on magnetic particles. Water Air Soil Pollut. 2021;232(12):1–18.
39. Mohanty SS, Kumar A. Biodegradation of Indanthrene Blue RS dye in immobilized continuous upflow packed bed bioreactor using corncob biochar. Sci Rep. 2021;11(1):13390.
40. Moosavi S, Lai CW, Gan S, Zamiri G, Akbarzadeh Pivehzhani O, Johan MR. Application of magnetic particles and activated carbon for dye removal from wastewater. ACS Omega. 2020;5(33):20684–97.
41. Moreno Castro DW, Franco Arias OO, Valenzuela Cobos JD, Prieto Sánchez D, Pimenteira C. Data mining to evaluate the effect of *Eichhornia crassipes* and *Lemna minor* in the phytoremediation of wastewater in the Canton of Milagro. Water. 2025;17(10):1551.
42. Mishra V, Mukherjee P, Bhattacharya S, Sharma RS. Innovative sustainable solutions for detoxifying textile industry effluents using advanced oxidation and biological methods. J Environ Manage. 2025;380:124804.
43. Neetha JN, Ujwal P, Sandesh K, Santhosh H, Girish K. Aerobic biodegradation of acid blue-9 dye by *Bacillus fermus* isolated from *Annona reticulata*. Environ Technol Innov. 2018;11:253–61.
44. Ong SA, Ho LN, Wong YS, Mohammad Pakri KA. Comparative study on biodegradation of mixed Remazol dyes between integrated anaerobic/aerobic and aerobic SBRs. Rend Lincei. 2017;28(3):497–501.
45. Pahlevanneshan Z, Deypour M, Kefayat A, Rafienia M, Sajkiewicz P, Neisiany R, et al. Polyurethane-nanolignin composite foam coated with propolis for wound dressing. Polymers. 2021;13(18):3191.
46. Partovinia A, Rasekh B. Review of immobilized microbial cell systems for bioremediation of petroleum hydrocarbons. Crit Rev Environ Sci Technol. 2018;48(1):1–38.
47. Peng FJ, Li S, You WD, Wang XY, Feng XJ, Yang B, et al. Toxicity assessment of wastewater using a battery of bioassays in two textile wastewater treatment plants from a large industrial park in Guangxi, Southwest China. Sci Total Environ. 2025;966:178766.
48. Rajendran R, Prabhavathi P, Karthiksundaram S, Pattabi S, Kumar SD, Santhanam P. Biodecolourization and bioremediation of denim wastewater by adapted bacterial consortium immobilized on polyurethane foam. Pol J Microbiol. 2015;64(4):329–38.
49. Rawat D, Sharma RS, Karmakar S, Arora LS, Mishra V. Ecotoxic potential of a presumably non-toxic azo dye. Ecotoxicol Environ Saf. 2018;148:528–37.
50. Ratusznei SM, Rodrigues JAD, Camargo EFMD, Zaiat M, Borzani W. Feasibility of a stirred anaerobic sequencing batch reactor containing immobilized biomass. Bioresour Technol. 2000;75(2):127–32.
51. Rede D, Santos LH, Ramos S, Oliva-Teles F, Antão C, Sousa SR, et al. Individual and mixture toxicity evaluation of pharmaceuticals to *Lactuca sativa*. Sci Total Environ. 2019;673:102–9.
52. Saleem, M. U., Khan, S. J., Shahzad, H. M. A., & Zeshan. (2022). Performance evaluation of integrated anaerobic and aerobic reactors for treatment of textile wastewater. International Journal of Environmental Science and Technology, 19(10), 10325–10336.
53. Suvaitha SP, Venkatachalam K. A study of kinetics, thermodynamics, and adsorption isotherms of food and textile dyes removal with SBA-15 supported polyaniline nanocomposite. Water Air Soil Pollut. 2023;234(4):264.
54. Swain G, Maurya KL, Sonwani RK, Singh RS, Jaiswal RP, Rai BN. Effect of mixing intensity on biodegradation of phenol in a moving bed biofilm reactor. Bioresour Technol. 2022;351:126921.
55. Takkar S, Tyagi B, Kumar N, Kumari T, Iqbal K, Varma A, et al. Biodegradation of methyl red dye by a novel *Zhihengliuella* sp. ISTPL4. Environ Technol Innov. 2022;26:102348.
56. Tizazu S, Tesfaye G, Wang A, Guadie A, Andualem B. Microbial diversity, transformation and toxicity of azo dye biodegradation using thermo-alkaliphilic microbial consortia. Heliyon. 2023. https://doi.org/10.1016/j.heliyon.2023.e16857.
57. Tomei MC, Pascual JS, Angelucci DM. Analysing performance of real textile wastewater bio-decolourization under different reaction environments. J Clean Prod. 2016;129:468–77.

58. Tripathi P, Tiwari S, Sonwani RK, Singh RS. Enhancing biofilm-mediated degradation of brilliant green dye in packed bed bioreactor. Process Saf Environ Prot. 2023;170:1228–39.
59. Veras ST, Rojas P, Florencio L, Kato MT, Sanz JL. 1,3-propanediol production from glycerol in polyurethane foam containing anaerobic reactors. Environ Sci Pollut Res. 2020;27(36):45662–74.
60. Wang W, Chang JS, Show KY, Lee DJ. Anaerobic recalcitrance in wastewater treatment: a review. Bioresour Technol. 2022;363:127920.
61. Wang X, Li X, Qiu J, Li Y. Microbial communities on different packing media in biofilter. Rev Compos Mater Avances. 2019. https://doi.org/10.18280/rcma.290306.
62. You SJ, Damodar RA, Hou SC. Degradation of reactive black 5 dye using anaerobic/aerobic membrane bioreactor and photochemical membrane reactor. J Hazard Mater. 2010;177:1112–8. https://doi.org/10.1016/j.jhazmat.2010.01.036.
63. Ziganshin A, Wintsche B, Seifert J, Carstensen M, Born J, Kleinsteuber S. Spatial separation of metabolic stages in a tube anaerobic baffled reactor. Appl Microbiol Biotechnol. 2019;103:3915–29.
64. Samuchiwal, S., Gola, D., & Malik, A. (2021). Decolourization of textile effluent using native microbial consortium enriched from textile industry effluent. Journal of Hazardous Materials, 402, 123835. https://doi.org/10.1016/j.jhazmat.2020.123835
65. Mendes, P. M., Ribeiro, J. A., Martins, G. A., Lucia, T., Araujo, T. R., Fuentes-Guevara, M. D., Corrêa, L. B., & Corrêa, É. K. (2021). Phytotoxicity test in check: Proposition of methodology for comparison of different method adaptations usually used worldwide. Journal of Environmental Management, 291, 112698.https://doi.org/10.1016/J.JENVMAN.2021.112698
66. Jafari, M., Desmond, P., van Loosdrecht, M. C. M., Derlon, N., Morgenroth, E., & Picioreanu, C. (2018). Effect of biofilm structural deformation on hydraulic resistance during ultrafiltration: A numerical and experimental study. Water Research, 145, 375–387. https://doi.org/10.1016/J.WATRES.2018.08.036
67. Carpio, A. R., Avallone, F., Ragni, D., Snellen, M., & Zwaag, S. van der. (2019). 3d-printed perforated trailing edges for broadband noise abatement. 25th AIAA/CEAS Aeroacoustics Conference, 2019.https://doi.org/10.2514/6.2019-2458
68. Žaltauskaitė, J., Sujetovienė, G., Čypaitė, A., & Aužbikavičiūtė, A. (2014). Lemna minor as a tool for wastewater toxicity assessment and pollutants removal agent. 9th International Conference on Environmental Engineering, ICEE 2014.https://doi.org/10.3846/ENVIRO.2014.104
69. Ravindran, B., Kumari, S. K. S., Stenstrom, T. A., & Bux, F. (2016). Evaluation of phytotoxicity effect on selected crops using treated and untreated wastewater from different configurative domestic wastewater plants. Environmental Technology (United Kingdom), 37(14), 1782–1789.https://doi.org/10.1080/09593330.2015.1132776

Publisher's Note